Potential applications of Species Distribution Modeling

Modeling the distribution of marine biodiversity under global climate change

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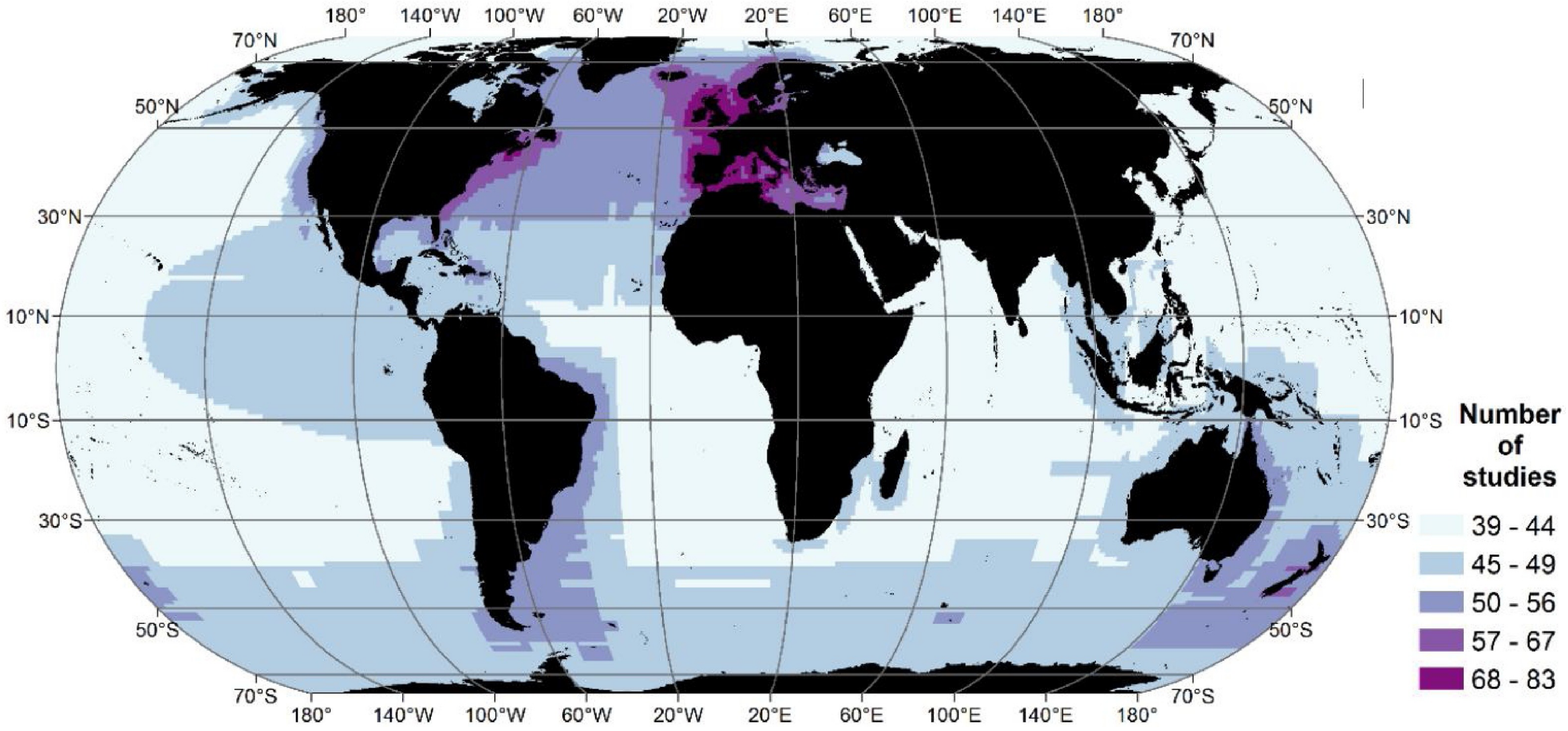

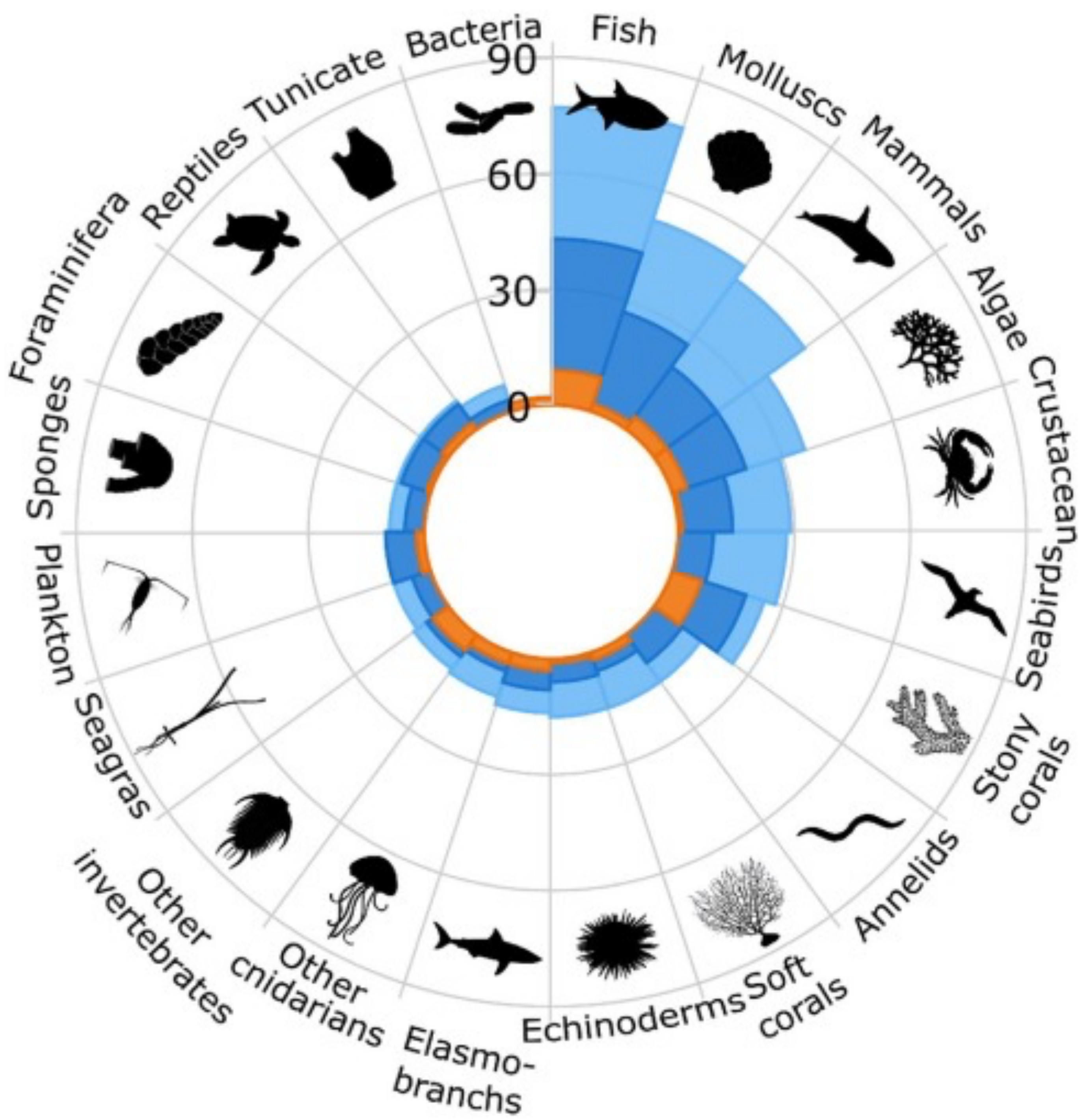

The use of SDMs have experienced an explosive growth.

Systematic review of 328 articles showed information **bias towards regions and groups**.

Most studies were based on **correlative approaches** focused on the **mechanisms underlying geographic ranges** (64%), **evaluate impacts of climate change** (19%) and develop **strategies for conservation** (11%).

Melo-Merino et al., 2020

Drivers and patterns of species distributions

Contents lists available at ScienceDirect

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ELSEVIER

Environmental drivers of rhodolith beds and epiphytes community along the South Western Atlantic coast

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A R T I C L E I N F O

Keywords:
Habitat-building
Marine ecology
Macroalgae
Temperature
Light
Nitrate

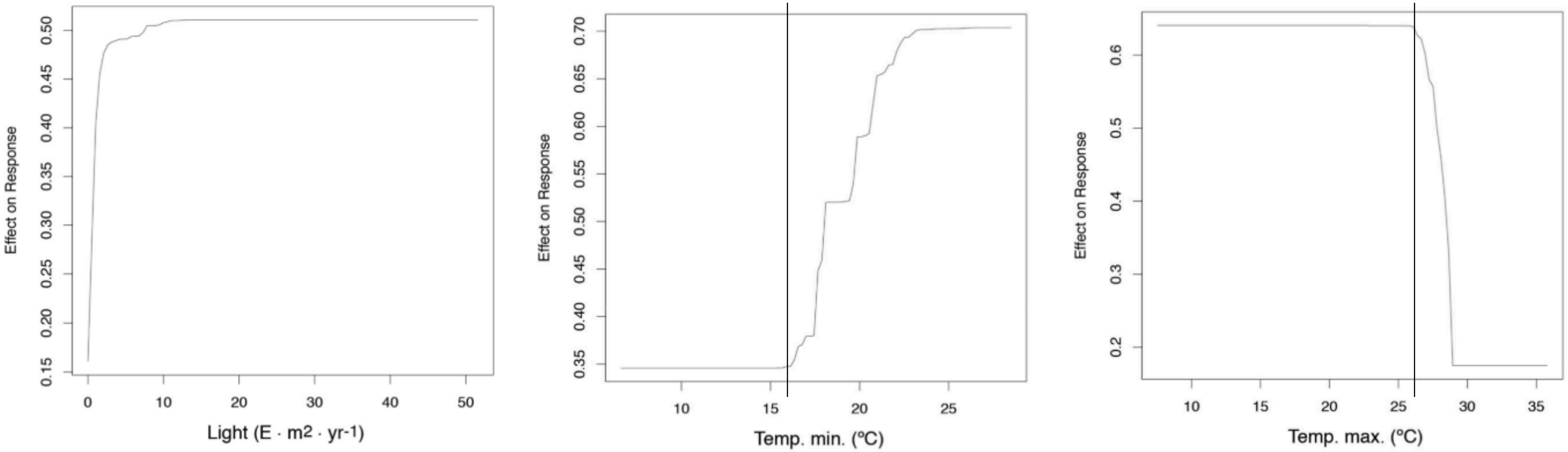
A B S T R A C T

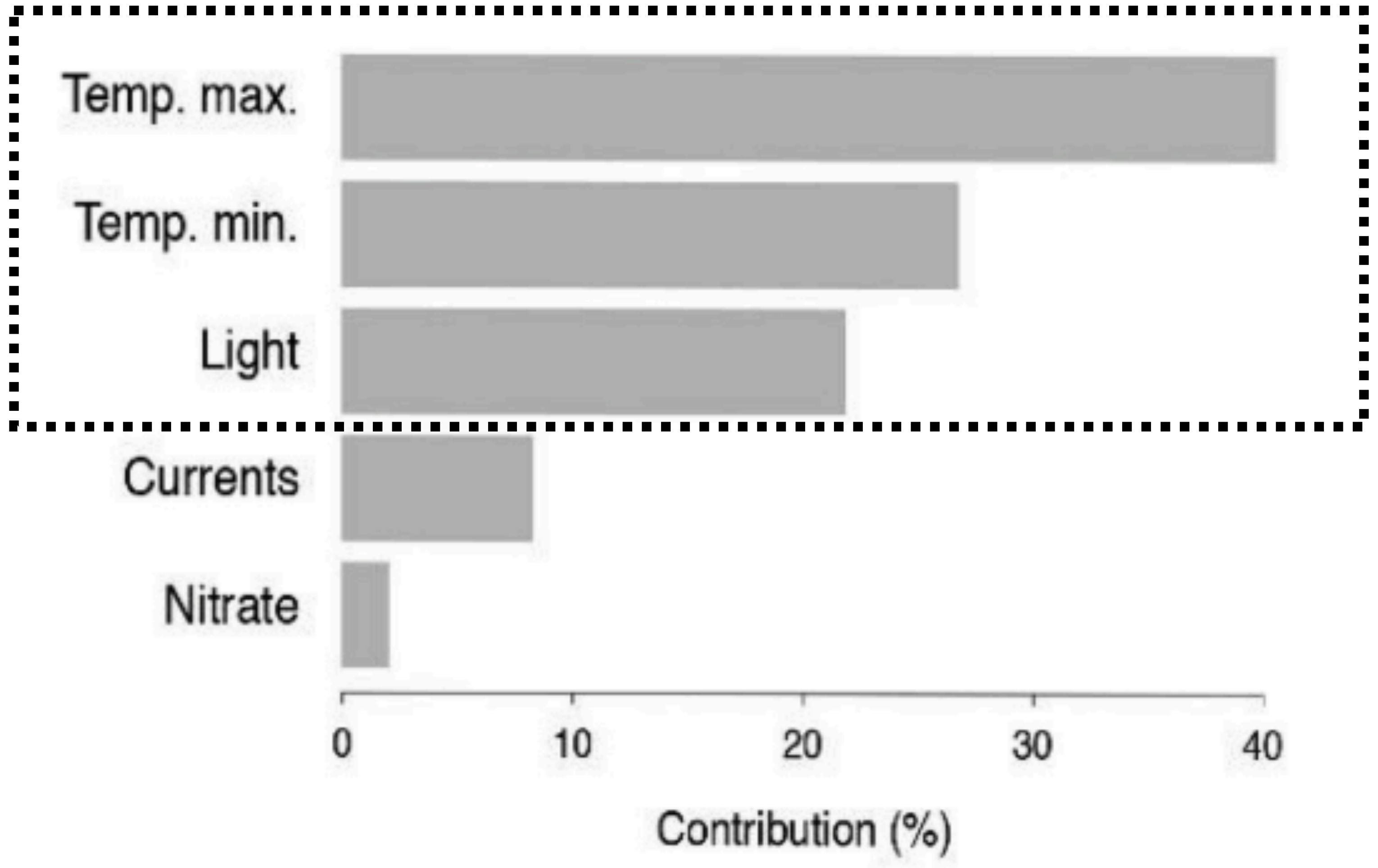
Environmental conditions shape the occurrence and abundance of habitat-building organisms at global scales. Rhodolith beds structure important hard substrate habitats for a large number of marine benthic organisms. These organisms can benefit local biodiversity levels, but also compete with rhodoliths for essential resources. Therefore, understanding the factors shaping the distribution of rhodoliths and their associated communities along entire distributional ranges is of much relevance for conservational biology, particularly in the scope of future environmental changes. Here we predict suitable habitat areas and identify the main environmental drivers of rhodoliths' variability and of associated epiphytes along a large-scale latitudinal gradient. Occurrence and abundance data were collected throughout the South-western Atlantic coast (SWA) and modelled against high resolution environmental predictors extracted from Bio-Oracle. The main drivers for rhodolith occurrence

Rationale: Rhodolith beds structure important habitats for a large number of organisms. However, their **deep and cryptic occurrence hinders knowledge about where they are, and about the factors shaping their distribution**.

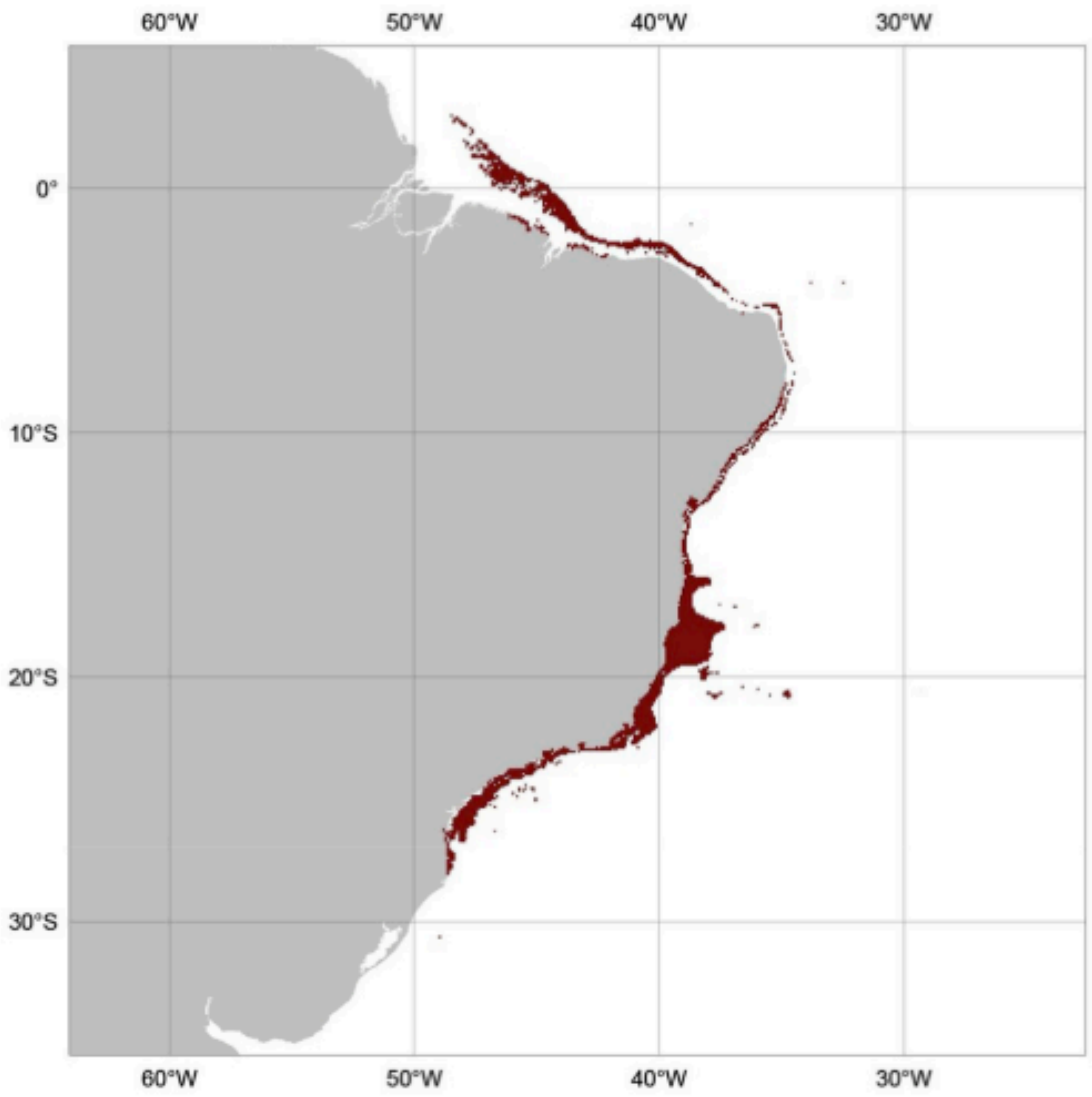
Goal: **identify the distribution drivers and patterns of rhodoliths**.

Methods: occurrence records collected throughout the South-western Atlantic coast and modeled against environmental predictors extracted from Bio-Oracle.





Results: main drivers for rhodolith distribution are light availability and temperature, with clear ecological responses supported by additional empirical studies.



Results: the Brazilian tropical-temperate region provides broad suitable areas for rhodoliths, decreasing toward extreme warm and cold regions.

Identify areas that should be prioritized for conservation

SCIENTIFIC REPORTS

OPEN

Overlooked habitat of a vulnerable gorgonian revealed in the Mediterranean and Eastern Atlantic by ecological niche modelling

Received: 21 December 2015
Accepted: 17 October 2016
Published: 14 November 2016

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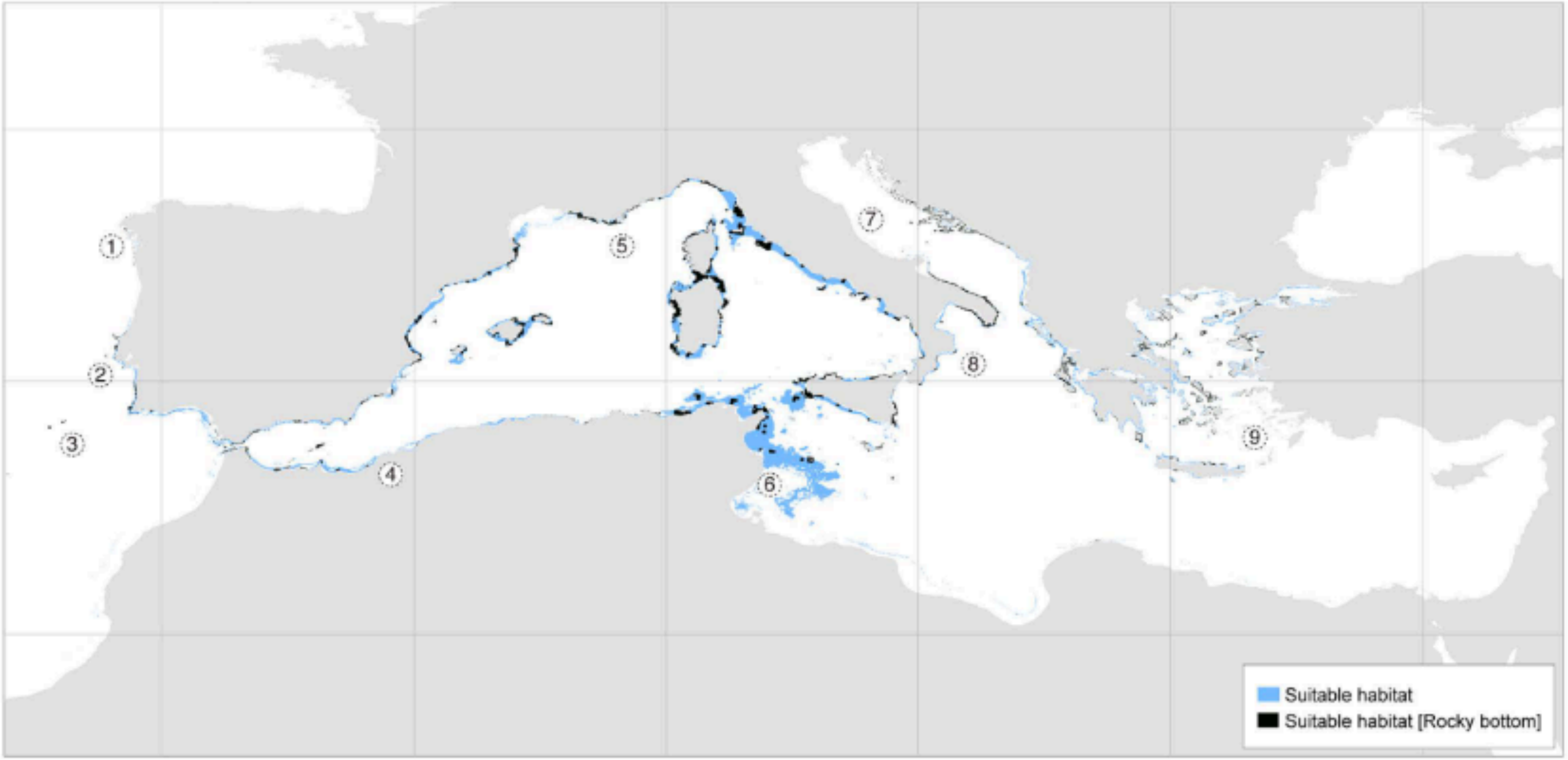
Factors shaping the distribution of mesophotic octocorals (30–200 m depth) remain poorly understood, potentially leaving overlooked coral areas, particularly near their bathymetric and geographic distributional limits. Yet, detailed knowledge about habitat requirements is crucial for conservation of sensitive gorgonians. Here we use Ecological Niche Modelling (ENM) relating thirteen environmental predictors and a highly comprehensive presence dataset, enhanced by SCUBA diving surveys, to investigate the suitable habitat of an important structuring species, *Paramuricea clavata*, throughout its distribution (Mediterranean and adjacent Atlantic). Models showed that temperature (11.5–25.5 °C) and slope are the most important predictors carving the niche of *P. clavata*. Prediction throughout the full distribution (TSS 0.9) included known locations of *P. clavata* alongside with previously unknown or unreported sites along the coast of Portugal and Africa, including seamounts. These predictions increase the understanding of the potential distribution for the northern Mediterranean and indicate

Rationale: **the distribution of corals remain poorly understood, potentially leaving overlooked distributions**, particularly in non-surveilled areas (e.g., deep unaccessible rocky reefs). **Knowledge about their distributions is crucial for conservation**.

Goal: **investigate the distribution of suitable habitats of an important structuring species (*Paramuricea sp.*) to inform conservation**.

Methods: occurrence records collected throughout the entire Mediterranean and Atlantic distribution and modeled against environmental predictors extracted from Bio-Oracle.



Results: predictive maps increased our **understanding in areas of the deep northern Mediterranean** (> 150m), and in **previously unknown habitats along Algeria, Alboran Sea and adjacent Atlantic;** encouraged deep exploration (NGS) to confirm its existence.

Map habitats of commercial species under climate change

Received: 9 November 2020 | Revised: 30 April 2021 | Accepted: 12 May 2021

DOI: 10.1111/geb.13327

RESEARCH ARTICLE

Global Ecology and Biogeography A Journal of Macroecology

WILEY

Biologically meaningful distribution models highlight the benefits of the Paris Agreement for demersal fishing targets in the North Atlantic Ocean

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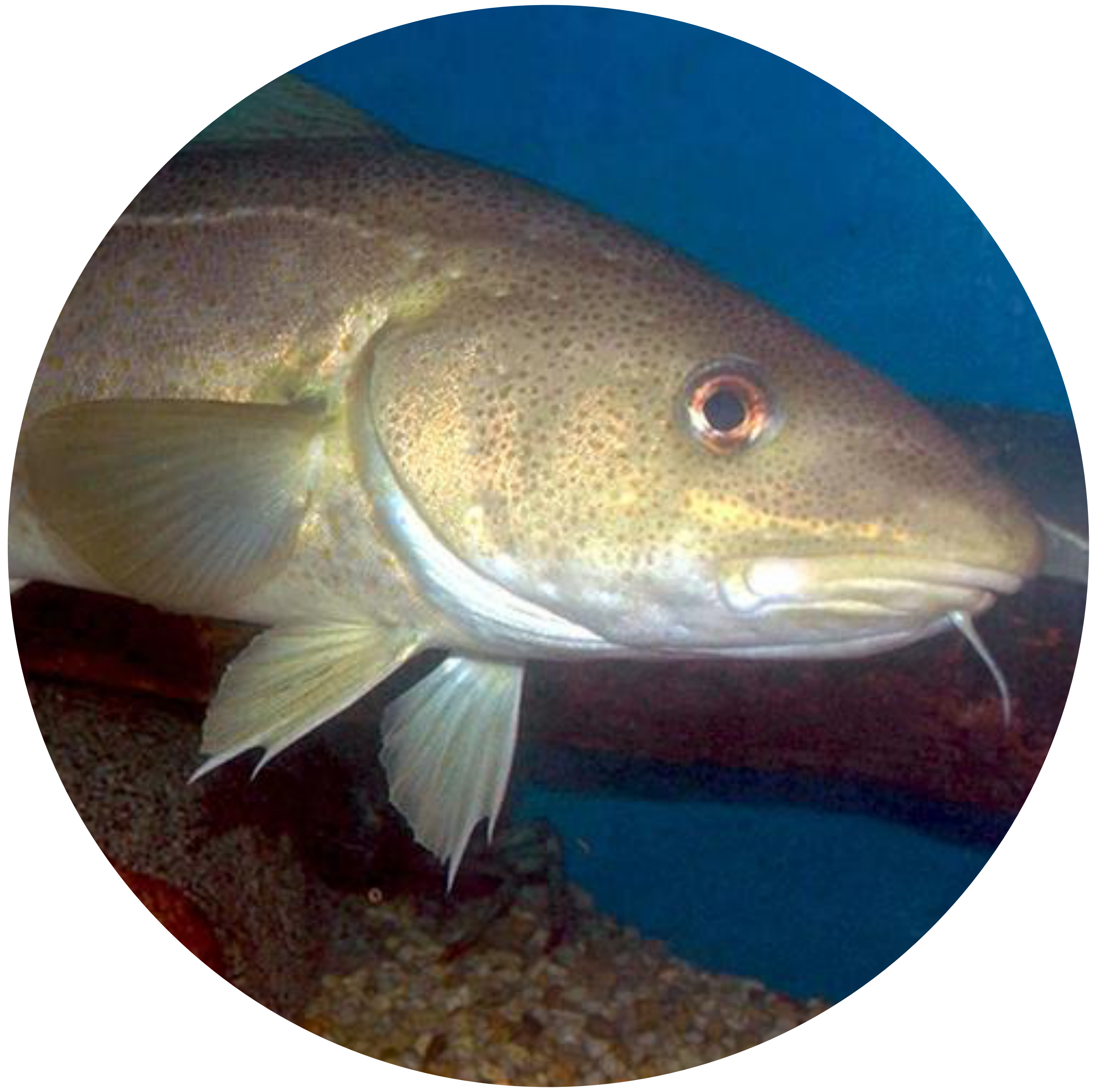
Abstract

Aim: With climate change challenging marine biodiversity and resource management, it is crucial to anticipate future latitudinal and depth shifts under contrasting global change scenarios to support policy-relevant biodiversity impact assessments [e.g., Intergovernmental Panel on Climate Change (IPCC)]. We aim to demonstrate the benefits of complying with the Paris Agreement (United Nations Framework Convention on Climate Change) and limiting environmental changes, by assessing future distributional shifts of 10 commercially important demersal fish species.

Location: Northern Atlantic Ocean.

Time period: Analyses of distributional shifts compared near present-day conditions



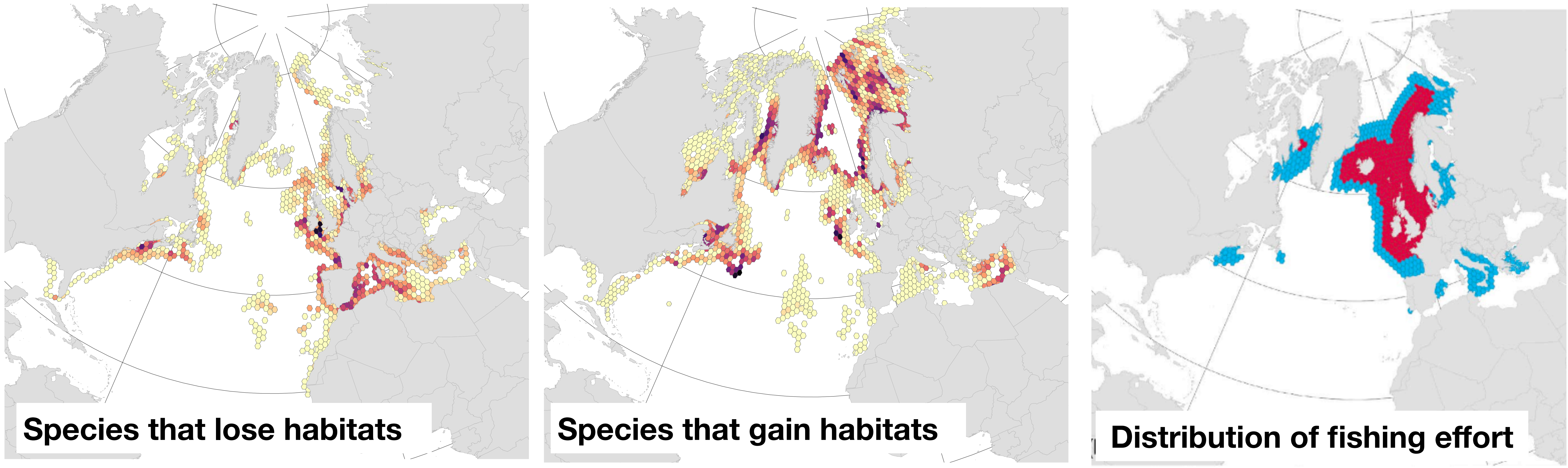
Rationale: climate change is expected to **shift the distribution of marine species, challenging resource management**. Demersal fish, which are key to commercial fisheries, are **vulnerable to shifts in temperature, oxygen, and productivity**.

Goal: assess future distribution shifts of 10 commercially important fish species and evaluate the benefits of limiting climate change under the Paris Agreement.

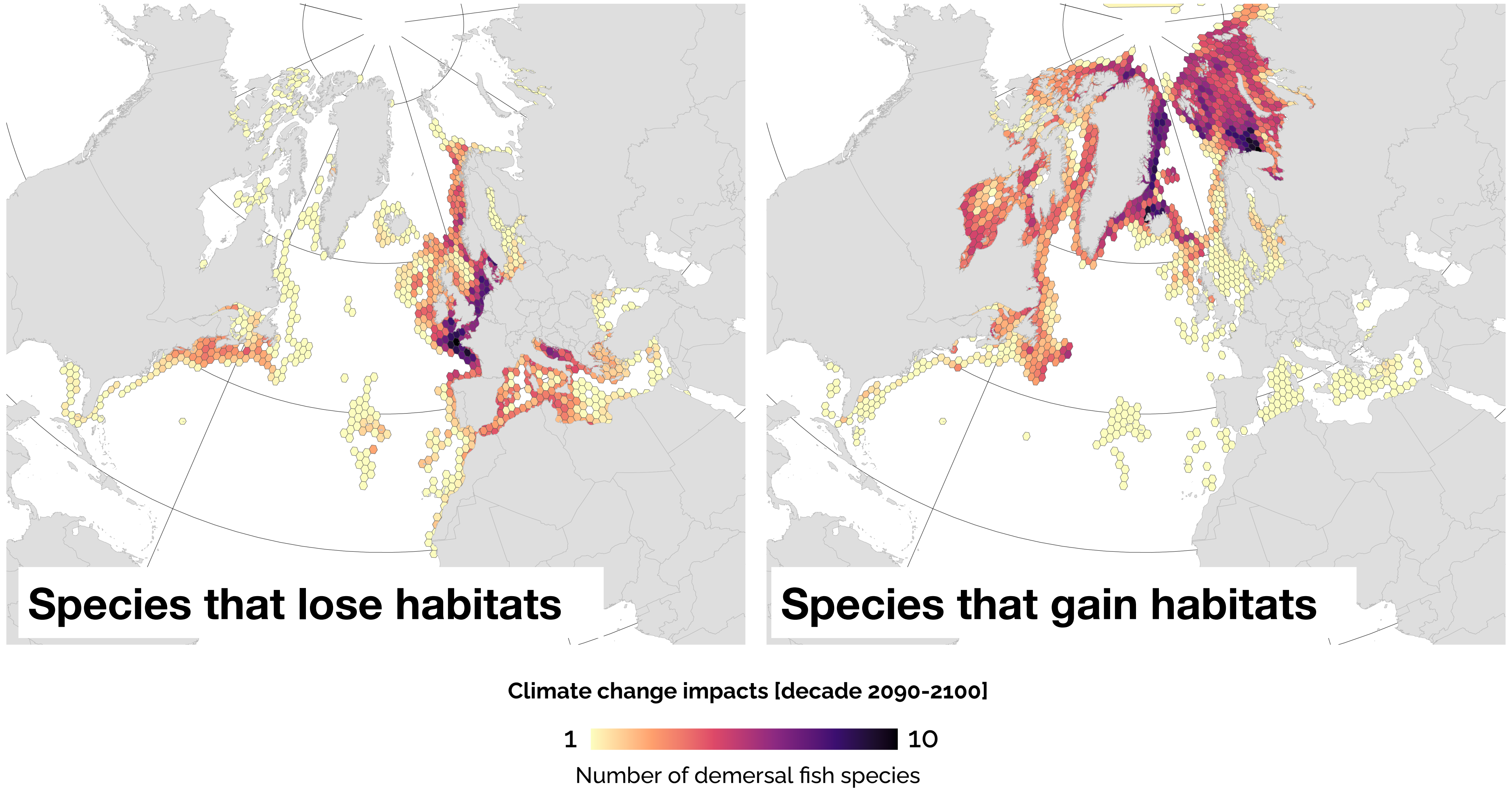
Methods: occurrence records compiled from literature and major databses, and relevant environmental predictors from Bio-Oracle.

biodiversityDS.

SSP1-1.9 (Paris agreement expectations)



SSP3-7.0 (High emissions)



biodiversityDS.

Evaluate new areas for invasive species

Science of the Total Environment 700 (2020) 134692

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Science of the Total Environment

journal homepage: www.elsevier.com/locate/scitotenv

How experimental physiology and ecological niche modelling can inform the management of marine bioinvasions?

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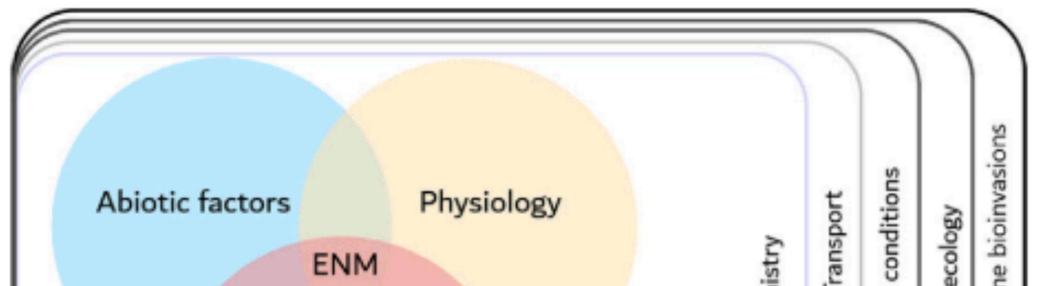
H I G H L I G H T S

- An Ecological Niche Model was built to explore the distribution of an invasive alga.
- Model results were crosschecked with ecophysiological evaluations.
- *G. turuturu* has higher habitat

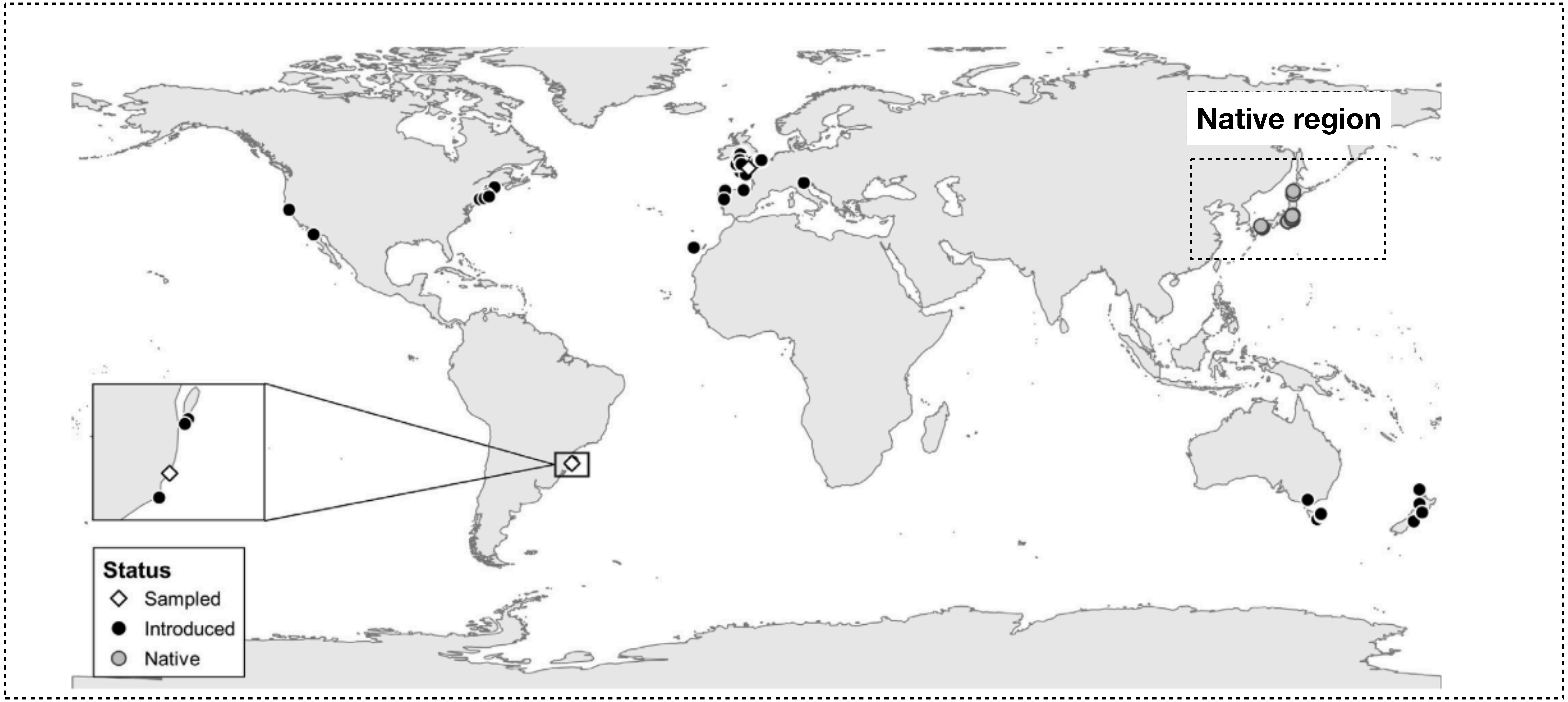
G R A P H I C A L A B S T R A C T

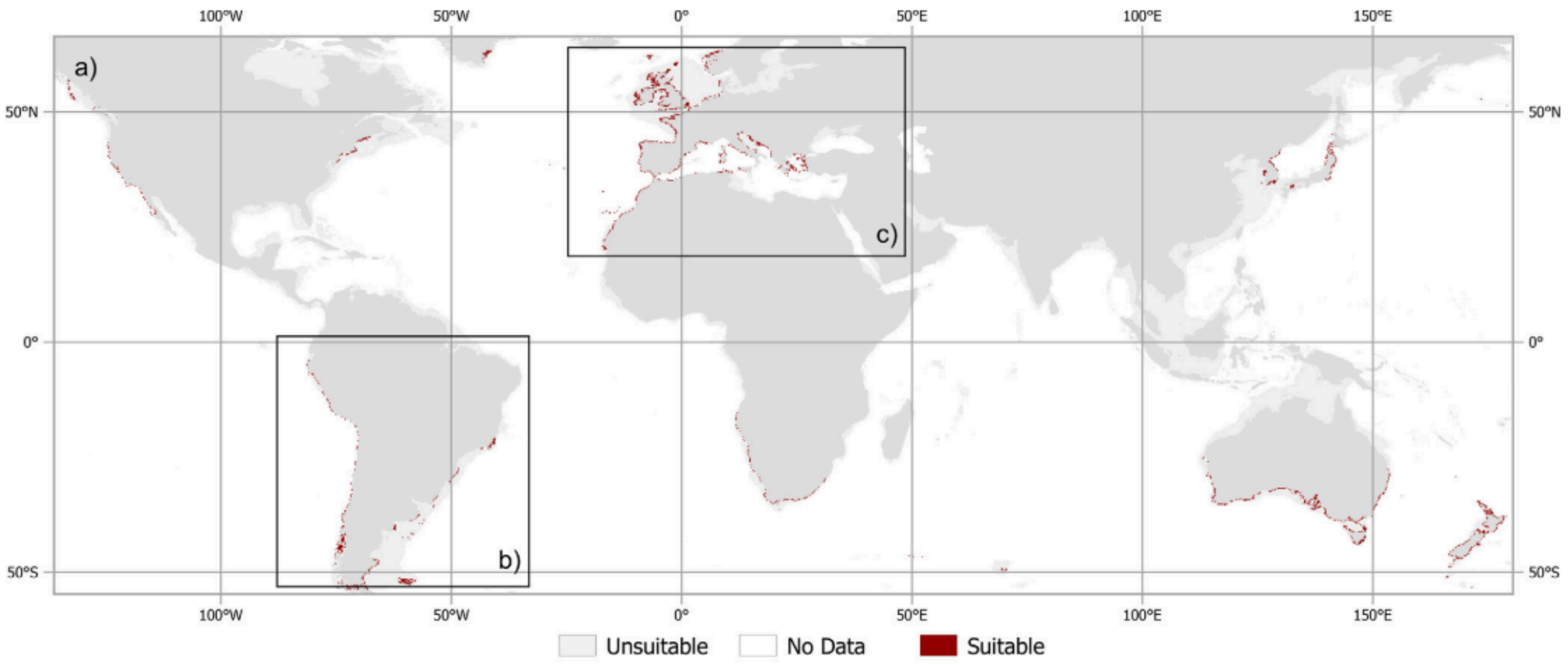


Rationale: information on habitats of species with high invasive potential is crucial for management decisions aiming to prevent their arrival and spread. *Grateloupia turuturu*, is one of the most harmful species, capable of causing biodiversity losses.

Goal: infer the potential global distribution to assess potential areas of new introductions.

Methods: records collected by field sampling and the available literature, and modeled with high resolution environmental predictors extracted from Bio-Oracle.


Native region
Status
Sampled
Introduced
Native



Grateloupia turuturu **habitat suitability**

Results: **high suitability in temperate and warm regions** around the world, with **focus on areas where this species still doesn't occur**. Management initiatives can mitigate anthropogenic transport and promote eradication, with **focus in the South America, South Africa, Southern and Western Australia, and New Zealand**.

References

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Koerich et al., 2020: https://doi.org/10.1016/j.scitotenv.2019.134692